


**UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE**

**DEPARTMENT OF COMPUTER SCIENCE**

**Object-Oriented Programming**

**Homework 5-Product Backlog**

**Group N°1**
**Project: Liquor Store System**
**Team Name: Logical Lords**

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**First Version Scope**

The first version will support the daily operations of one liquor store through a console application. Information will be entered manually and stored in local JSON files.

Sales will be recorded as cash or credit transactions. The owner and employee will be identified as the responsible sellers. Discounts will be entered as percentages applied to the total sale amount.

| Story ID | User Story | Task Name | Task Owner | Priority |
|---|---|---|---|---|
| US001 | As an owner, I want to register products, record stock entries and exits, and view available quantities so that I can maintain accurate inventory records. | Develop product and stock management | Acurio | High |
| US002 | As an owner, I want to set a minimum stock level for each product and view low-stock alerts so that I can identify products that require restocking. | Implement low-stock alerts | Acurio | High |
| US003 | As an owner, I want to register suppliers, record purchase orders, and confirm deliveries so that I can track pending orders and update inventory when products are received. | Develop supplier and purchase order management | Ayala | High |
| US004 | As an owner, I want each sale to record its products, quantities, prices, total amount, date, and responsible seller so that I can review sales made by the owner and the employee. | Develop sales registration and seller tracking | Armas | High |
| US005 | As an owner, I want to record customer credit purchases and payments so that I can view payment history and check each customer's outstanding balance. | Develop customer credit and payment management | Armas | High |
| US006 | As an owner, I want to record the opening cash balance, cash receipts, and expenses and compare the expected closing balance with the amount counted so that I can identify daily cash differences. | Develop daily cash reconciliation | Acurio | High |
| US007 | As an owner, I want to calculate gross profit and profit margins from purchase costs and actual selling prices so that I can check the expected 20% margin and understand how discounts affect profit. | Develop gross profit and margin calculations | Armas | High |
| US008 | As a cashier, I want to enter a discount percentage authorized by the owner and have the system calculate the final sale amount so that customers are charged correctly. | Implement percentage discounts | Ayala | Medium |
| US009 | As an owner, I want to view products ranked by units sold during a selected period so that I can identify best-selling products and make restocking decisions. | Develop best-selling products report | Ayala | Medium |

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| US010 | As a cashier, I want an automatic age-verification reminder before completing each sale so that I remember to check the customer's age. | Implement age-verification reminder | Ayala | Medium |

**Acceptance Criteria**

US001 Product and Stock Management

- Products are registered with a unique code, name, purchase cost per unit, labeled selling price, and available quantity.

-Initial stock must be a whole number equal to or greater than zero. Purchase costs cannot be negative, and labeled selling prices must be greater than zero.

-The system allows the user to update product information and record stock entries and exits.

-Stock movements require positive quantities. An exit cannot exceed the available stock.

-Product quantities remain available after the application is restarted.

US002 Low Stock Alerts

-Each product has a minimum stock level equal to or greater than zero.

-The console identifies products whose available quantity is equal to or below their minimum level.

-Alerts reflect the updated quantities after stock changes.

US003 Suppliers and Purchase Orders

-Supplier records include an identifier, name, and contact information.

-Purchase orders identify the supplier, products, quantities, and status.

-Orders begin as Pending and become Received when the delivery is confirmed.

-Confirming receipt increases stock. The same order cannot increase stock more than once.

US004 Sales Registration

-A sale contains at least one product, positive quantities, a date, and an identified seller.

-The system calculates the subtotal and final total using the quantities, prices, and applicable discount.

-A sale cannot be completed if the requested quantities exceed available stock.

-The final sale amount must be greater than zero after applying the discount and rounding to two decimal places.

-Each completed sale stores the purchase cost per unit, selling price per unit, quantities, and discount used at the time of the transaction.

-Later changes to product costs or prices do not modify previously recorded sales.

-Completing a sale records it and reduces inventory. Canceling it leaves inventory and financial balances unchanged.

US005 Customer Credit and Payments

-Credit sales are associated with an identified customer and increase that customer's outstanding balance.

-The system records each payment with its amount and date and reduces the outstanding balance.

-Payments must be greater than zero and cannot exceed the outstanding balance.

-The owner can consult the customer's credit purchases, payment history, and remaining balance.

US006 Daily Cash Reconciliation

-The user records an opening cash balance and cash expenses for the selected day.

-Opening cash and counted cash cannot be negative. Recorded expenses must be greater than zero.

-Expected closing cash equals opening cash plus cash sales and customer debt payments received on the selected day, minus cash expenses paid on that day.

-Unpaid credit sales do not increase the cash balance.

-The cash difference equals counted cash minus expected closing cash. A positive difference indicates a surplus, and a negative difference indicates a shortage.

-The report displays the selected date, expected cash, counted cash, and cash difference.

US007 Gross Profit and Margin

- Purchase costs and labeled selling prices are entered per product unit.

-Gross profit equals the actual sale amount after discounts minus the total purchase cost of the units sold.

-Calculations use the costs, prices, quantities, and discounts stored with the sale. Later product updates do not modify historical gross profit.

-Gross profit margin equals gross profit divided by the actual sale amount, multiplied by 100. A margin calculation with a zero or negative sale amount is rejected.

-A purchase cost of $8 and a selling price of $10 produce $2 of gross profit and a 20% margin.

-The system displays the calculated margin without automatically changing the labeled selling price.

US008 Percentage Discounts

-The cashier can enter a discount percentage from 0% up to, but not including, 100%.

-A 0% discount leaves the sale amount unchanged.

-The system calculates the discount amount and final total and records the discount with the sale.

-The discounted amount is used consistently in payment and gross profit calculations.

US009 Best Selling Products

-The user selects the start and end dates of the reporting period.

-The start date must be earlier than or equal to the end date.

-The report includes completed sales made on both the start date and the end date, as well as -the dates between them.

-The report totals the units sold for each product in completed sales, including credit sales.

-Products are listed from highest to lowest quantity sold.

-If the period contains no sales, the console displays an informative message.

US010 Age Verification Reminder

-The console displays the reminder before a sale is completed.

-The cashier confirms that the age check has been performed or cancels the sale.

-Canceling at this step leaves stock, customer debt, and cash balances unchanged.

**Persistence and Validation**

Successful changes to products, suppliers, orders, sales, customer accounts, payments, and cash records are saved in local JSON files and restored when the application starts.

Invalid input must produce a clear message and allow correction without partially updating the records. Monetary amounts are displayed with two decimal places.